

CBTC Migration Checklist

What has to be true at each stage of a brownfield conversion, from the first business case to the ninetieth day of revenue service — plus the four cutover strategies and where each one fits.

1 · FEASIBILITY & BUSINESS CASE

- ☐ Strategic trigger named explicitly: capacity ceiling, asset obsolescence, or a reliability floor — not "modernization".
- ☐ Scope options costed side by side, including the do-nothing and the conventional-resignaling cases.
- ☐ Ridership and capacity forecast tied to the headway the CBTC design will actually deliver in mixed mode, not at maturity.
- ☐ FTA Capital Investment Grant implications assessed before the scope is frozen.
- ☐ Whole-life cost modeled, including the software and radio refresh cycles a legacy estate does not have.

2 · PROCUREMENT

- ☐ Delivery method chosen and justified; bundled vs unbundled scope decided deliberately.
- ☐ Performance-based specification cites IEEE 1474 for performance and IEC 62290 for functional architecture and GoA.
- ☐ Interface Control Document scope written into the RFP: message formats, latency budgets, failure modes, acceptance criteria — per interface.
- ☐ Buy America compliance confirmed against each bidder's actual US footprint.
- ☐ Evaluation scoring published; performance guarantees and liquidated damages structured before award.
- ☐ Data rights, spares terms and long-term support priced now, not at the first change order.

3 · SYSTEMS ENGINEERING

- ☐ V-model established with verification planned against every requirement at the matching level.
- ☐ Requirements under configuration management from day one; traceability maintained, not reconstructed.
- ☐ Architecture allocation explicit about which functions are safety-critical and which are not.
- ☐ Owner's engineer in place with the standing to reject vendor interpretations.
- ☐ Audit of every existing system the CBTC must touch — interlocking, power, SCADA, PSD, PIS, depot — completed in the concept phase.

4 · SAFETY ASSURANCE

- ☐ Hazard analysis started early enough to change the design rather than document it.
- ☐ SIL allocation and tolerable hazard rates agreed and traceable to the architecture.
- ☐ Independent Safety Assessor appointed with genuine independence from the delivery team.
- ☐ State Safety Oversight and FTA pathway mapped, with submission dates on the master schedule.
- ☐ Safety case covers degraded modes and every train-type combination that will share track.

5 · CUTOVER

- ☐ Strategy selected against topology, fleet size, shutdown tolerance and rollback appetite (see matrix below).
- ☐ Method of Procedure written at 60–100 pages with flowcharts, responsibility matrix and escalation criteria.
- ☐ Rollback criteria pre-defined and objective — decided before the window, not inside it.
- ☐ Full cutover rehearsal 4–6 weeks prior with the same people, equipment and timeline.
- ☐ War room staffed: system integration lead, operations lead, safety lead / **ISA with authority to stop**, vendor engineering, interlocking specialist, communications coordinator.
- ☐ Mixed-mode capacity expectation communicated to stakeholders in advance and in writing.

6 · FIRST 90 DAYS

- ☐ Reduced schedule for burn-in; trains introduced in stages rather than all at once.
- ☐ Vendor engineering on site, not on call.
- ☐ Incident root-cause analysis turned around in days, with fixes tracked to closure.
- ☐ Capacity ramped only as evidence accumulates — the target headway is an outcome, not a launch commitment.

Cutover strategy decision matrix

Strategy	Best for	Shutdown	Mixed mode	Rollback risk	US examples
Big bang	Greenfield; simple topology	3–7 days	None	Moderate–high	Honolulu HART; Boston segments
Segment-phased	Urban brownfield; complex topology	3–4 days each	6–12 months	Low	NYC L Line; NYC 7 Line
Fleet-phased	Large fleet; staged retrofit budget	Incremental	18–36 months	Moderate	BART TCMP; Singapore MRT
Function-phased	Phased GoA roadmap and funding	Varies by phase	0–24+ months	Low per phase	BART TCMP; future WMATA plans

Mixed mode is a capacity decision, not a schedule detail. While CBTC and legacy trains share track, the least-capable train type governs headway: if legacy holds three minutes, so does everything else. A line engineered for 30 tph will run near 20 tph until the line or the fleet is fully converted. Budget the mixed-mode window as a degraded-capacity period and say so publicly before it starts.

Adapted from *Communications-Based Train Control*, Volume 1, Chapter 12, by Chunjun (Francisco) Wang. The cutover decision matrix is Table 12.7; strategy examples reflect documented US brownfield practice. Durations assume a mid-size urban line and compress substantially on greenfield.

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